SIEM (Security Information and Event Management)

SIEM (Security Information and Event Management) is a cybersecurity solution that collects, stores, analyzes, and correlates security logs and events from various devices and applications to detect threats and support incident response.

Definition

SIEM combines two technologies:

1. **SIM (Security Information Management)** – Collects and stores log data for analysis and compliance.
2. **SEM (Security Event Management)** – Monitors events in real time and generates alerts.

Thus,

SIEM = SIM + SEM

Architecture of SIEM

```
+----------------+
| Network Devices|
+----------------+
        |
+----------------+
| Firewalls      |
+----------------+
        |
+----------------+
| Servers        |
+----------------+
        |
+----------------+
| Endpoints      |
+----------------+
```

```
        |
+-----------------+
| Applications    |
+-----------------+
        |
        v
+-----------------+
| Log Collection |
+-----------------+
        |
        v
+-----------------+
| Normalization   |
+-----------------+
        |
        v
+-----------------+
| Correlation     |
+-----------------+
        |
        v
+-----------------+
| Alert Engine    |
+-----------------+
        |
        v
+-----------------+
| Dashboard       |
+-----------------+
        |
        v
+-----------------+
| Incident Response|
+-----------------+
```

Working of SIEM

Step 1: Data Collection

SIEM gathers logs from:

- Firewalls
- Routers
- Switches
- Servers
- Windows Event Logs
- Linux Syslogs
- Antivirus
- IDS/IPS
- Cloud services
- Databases
- Applications

Step 2: Log Normalization

Different devices produce logs in different formats.

Example:

Firewall:

```
Source IP: 10.1.1.5
```

Linux:

```
SRC=10.1.1.5
```

Windows:

```
Client Address: 10.1.1.5
```

SIEM converts them into a common format.

Step 3: Correlation

SIEM correlates events from multiple sources.

Example:

1. Multiple failed login attempts.
2. Successful login.
3. Privilege escalation.
4. Large data transfer.

SIEM identifies this sequence as a possible attack.

Step 4: Alert Generation

If predefined rules are matched:

- Email alerts
- SMS
- Dashboard notifications
- Ticket creation

Step 5: Incident Investigation

Security analysts:

- Examine logs.
- Trace attacker activity.
- Determine affected systems.
- Initiate containment and recovery.

Components of SIEM

Component	Function
Log Collector	Collects logs
Parser	Extracts information
Normalizer	Standardizes data
Correlation Engine	Detects attack patterns
Database	Stores logs
Dashboard	Displays events
Alert System	Sends notifications
Reporting Module	Generates compliance reports

Log Sources

SIEM collects logs from:

Network

- Firewalls
- Routers
- Switches

Security

- IDS
- IPS
- Antivirus
- EDR

Servers

- Linux
- Windows

Applications

- Web servers
- Mail servers
- Database servers

Cloud

- AWS
- Azure
- Google Cloud

Correlation Rules

Example:

Brute Force

```
IF
10 failed logins
within 2 minutes
from same IP

THEN
Generate alert
```

Port Scan

```
IF
One IP scans
100 ports
within 30 seconds

THEN
Alert
```

Malware

```
IF
Antivirus detects malware
AND
Host contacts suspicious IP

THEN
High severity alert
```

Features

- Centralized log management.
- Real-time monitoring.
- Event correlation.
- Threat detection.
- Alert generation.
- Incident response support.
- Compliance reporting.

- Forensic investigation.
- Dashboards and visualization.

Advantages

✅ Centralized logging.

✅ Faster threat detection.

✅ Real-time alerts.

✅ Compliance support.

✅ Reduced investigation time.

✅ Improved security visibility.

Disadvantages

❌ Expensive deployment.

❌ Requires skilled analysts.

❌ Large storage requirements.

❌ False positives.

❌ Complex rule tuning.

Applications

- Banking
- Government
- Healthcare
- Telecom
- Data centers
- Cloud environments
- Educational institutions

- Critical infrastructure

Common SIEM Tools

- Splunk Enterprise Security
- IBM QRadar
- ArcSight
- LogRhythm
- Microsoft Sentinel
- Wazuh
- OSSIM

SIEM vs IDS vs IPS

Feature	SIEM	IDS	IPS
Detect threats	Yes	Yes	Yes
Block attacks	Usually No	No	Yes
Log collection	Yes	Limited	Limited
Event correlation	Yes	No	No
Centralized management	Yes	No	No
Real-time alerts	Yes	Yes	Yes

SIEM Use Case Example

An attacker tries to compromise a server:

1. Firewall logs repeated connection attempts.
2. IDS detects suspicious traffic.
3. Server logs multiple failed logins.
4. Antivirus detects malicious execution.
5. SIEM correlates all these events.
6. SIEM raises a **high-priority alert** and notifies the security team.

SIEM (Security Information and Event Management) is a cybersecurity solution that collects, normalizes, stores, and analyzes security logs from multiple sources. It uses event correlation to detect suspicious activities and generate real-time alerts, helping organizations monitor security, investigate incidents, and meet compliance requirements. Its main functions include log management, threat detection, event correlation, alerting, incident investigation, and reporting. Popular SIEM tools include Splunk, IBM QRadar, Microsoft Sentinel, Wazuh, and OSSIM.